▶ HEMATOLOGY AND ONCOLOGY—EMBRYOLOGY

Fetal erythropoiesis

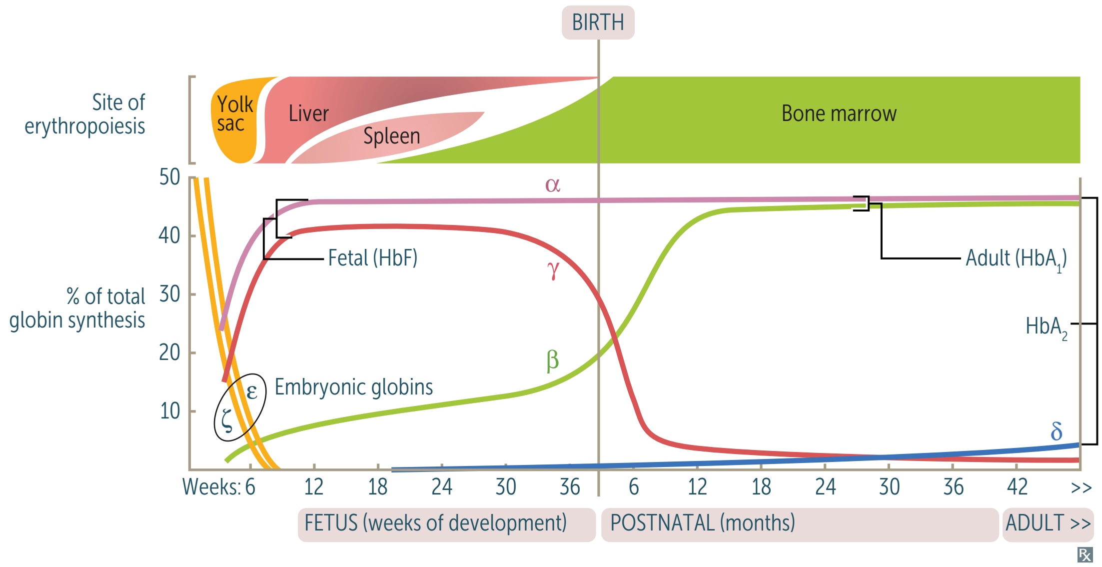
Fetal erythropoiesis occurs in:
- Yolk sac (3–8 weeks)
- Liver (6 weeks–birth)
- Spleen (10–28 weeks)
- Bone marrow (18 weeks to adult)

Young liver synthesizes blood.

Hemoglobin development

Embryonic globins: ζ and ε.
Fetal hemoglobin (HbF) = $\alpha_2\gamma_2$.
Adult hemoglobin (HbA_1) = $\alpha_2\beta_2$.
HbF has higher affinity for O_2 because it binds to 2,3-BPG with relatively lower affinity, allowing HbF to capture O_2 from maternal hemoglobin (HbA_1 and HbA_2) across the placenta. HbA_2 ($\alpha_2\delta_2$) is a form of adult hemoglobin present in small amounts.

From fetal to adult hemoglobin:
Alpha always; gamma goes, becomes beta.



Blood groups

	ABO classification				Rh classification	
	A	**B**	**AB**	**O**	**Rh⊕**	**Rh⊖**
RBC type	A	B	AB	O		
Group antigens on RBC surface	A	B	A & B	None	Rh (D)	None
Antibodies in plasma	Anti-B IgM	Anti-A IgM	None	Anti-A Anti-B IgG, IgM	None	Anti-D IgG
Clinical relevance Compatible RBC types to receive	A, O	B, O	AB, A, B, O	O	Rh⊕ , Rh⊖	Rh⊖
Compatible RBC types to donate to	A, AB	B, AB	AB	A, B, AB, O	Rh⊕	Rh⊕ , Rh⊖

Hemolytic disease of the fetus and newborn

Also called erythroblastosis fetalis. Most commonly involves the antigens of the major blood groups (eg, Rh and ABO), but minor blood group incompatibilities (eg, Kell) can also result in disease, ranging from mild to severe.

	ABO hemolytic disease	Rh hemolytic disease
INTERACTION	Type O pregnant patient; type A or B fetus.	Rh ⊖ pregnant patient; Rh ⊕ fetus.
MECHANISM	Preexisting pregnant patient anti-A and/or anti-B IgG antibodies cross the placenta → attack fetal and newborn RBCs → hemolysis.	First pregnancy: patient exposed to fetal blood (often during delivery) → formation of maternal anti-D IgG. Subsequent pregnancies: anti-D IgG crosses placenta → attacks fetal and newborn RBCs → hemolysis.
PRESENTATION	Mild jaundice in the neonate within 24 hours of birth. Unlike Rh hemolytic disease, can occur in firstborn babies and is usually less severe.	Hydrops fetalis, jaundice shortly after birth, kernicterus.
TREATMENT/PREVENTION	Treatment: phototherapy or exchange transfusion.	Prevent by administration of anti-D IgG to Rh ⊖ pregnant patients during third trimester and early postpartum period as well as in cases of ectopic pregnancy, miscarriage, abdominal trauma, and antepartum hemorrhage (if fetus Rh ⊕). Prevents maternal anti-D IgG production.